

Not all prices disinflate alike: disentangling the dynamics of sticky and flexible-price items

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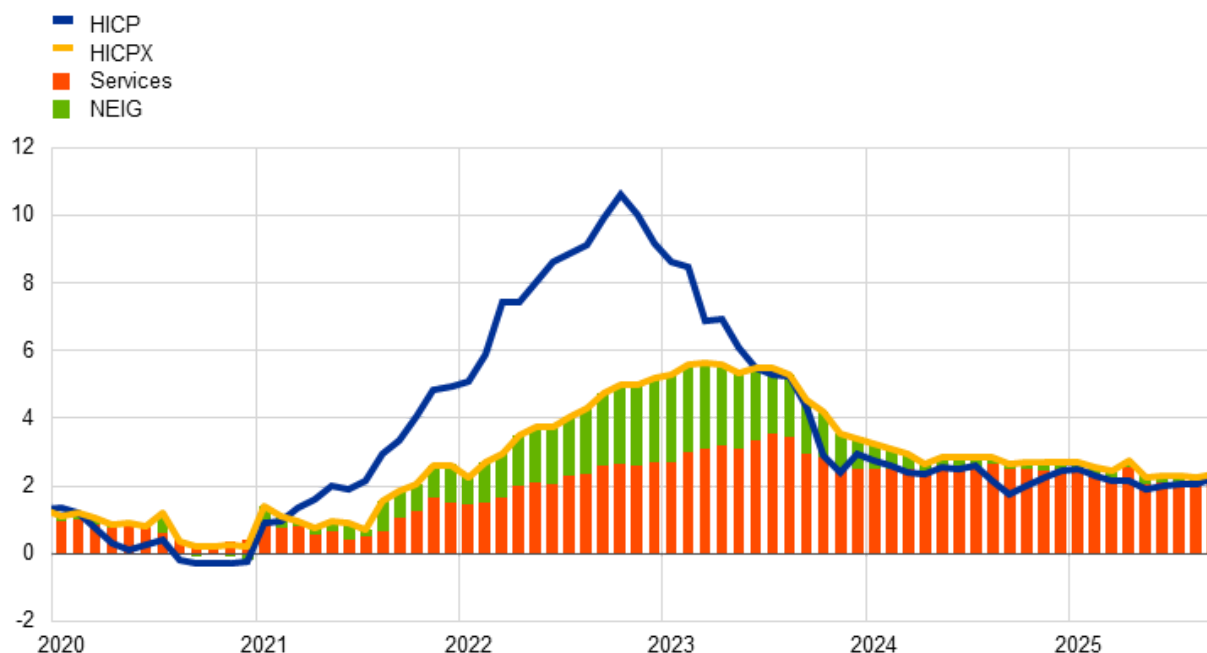
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Core inflation in the euro area has declined markedly since its 2023 peak but remains elevated, mainly due to persistent services price inflation. The annual rate of change in the Harmonised Index of Consumer Prices excluding energy and food (HICPX) has fallen substantially, from 5.7% in March 2023 to 2.4% in September 2025. However, it has stayed significantly above headline inflation for more than two years, underscoring its stickiness (Chart A). Within the HICPX, services inflation remains elevated, whereas non-energy industrial goods (NEIG) inflation has already returned to its pre-pandemic average. This box draws on micro price evidence on the frequency of price adjustments to disentangle the roles of sticky and flexible-price items in shaping recent disinflation dynamics in the HICPX.

Chart A

Developments in HICP and HICPX inflation

(annual percentage changes and percentage point contributions)



Sources: Eurostat and ECB staff calculations.

Note: The latest observations are for September 2025.

Micro price data allow inflation subcomponents to be divided into sticky and flexible items according to the underlying frequency of their price changes.

Following Bryan and Meyer (2010) for the United States, a sticky-price indicator should capture more persistent and generalised developments across prices than its flexible-price counterpart, as firms with infrequent price changes may place greater weight on expected future price developments when resetting prices.^[1] We thus draw on recent micro price evidence from Gautier et al. (2024) to construct measures of sticky and flexible core items for the euro area. For the low-inflation period of 2011-19, the authors report product-level frequencies of price adjustments for 11 euro area countries, covering about 60% of the euro area HICP basket.^[2] Our analysis for the euro area builds on 72 HICPX items according to the Classification of Individual Consumption according to Purpose (COICOP) at the four-digit level. Using cross-sectional frequency statistics (averaged over the period 2011-19) from Gautier et al. (2024), we classify an item as sticky if the frequency of its price changes is at or below the median frequency across all HICPX items. For the euro area, this median amounts to 7.3%, implying that in a typical month only 7.3% of product prices in the HICPX basket change relative to the previous month. Based on this criterion, 39 HICPX items are classified as sticky and the remaining 33 as flexible.^[3] Keeping this composition constant over time, the corresponding indices are then derived by aggregating sticky and flexible core items using the HICP's chain-linking and annual weighting method.

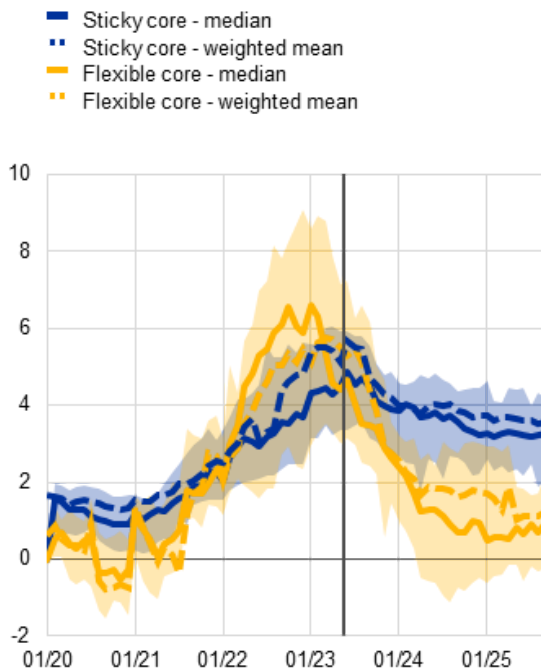
Sticky core inflation in the euro area has eased only gradually, while flexible core inflation has returned closer to its pre-pandemic average. Sticky core inflation, as measured by the weighted mean of sticky HICP items, has remained relatively persistent in recent months, hovering around 3.6% (Chart B, panel a). By contrast, flexible core inflation – which is more volatile and rose strongly during the inflation surge – started to decline earlier and has now fallen back closer to its pre-pandemic level, standing at 1.2% in September 2025. Flexible core items also displayed a higher degree of frontloading in price adjustments: for this category, 85% of items recorded their inflation peak before March 2023, which is when HICPX inflation reached its maximum. In comparison, only 41% of sticky core items had already peaked by that time.

Chart B

Price developments of sticky and flexible HICPX items

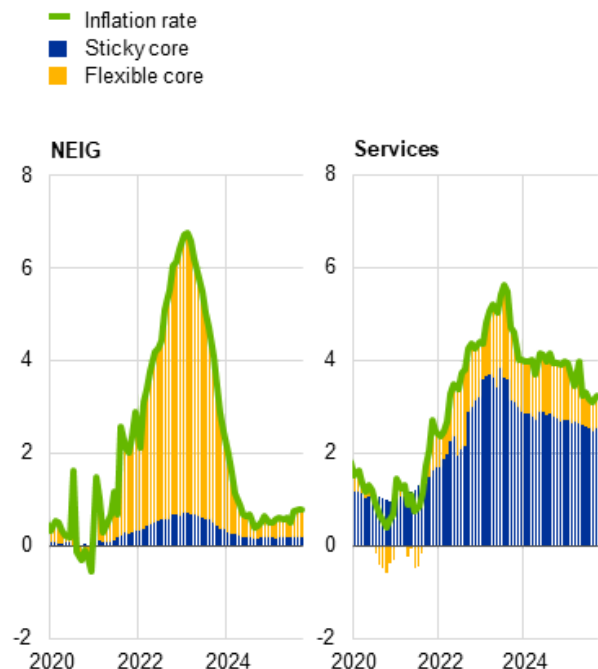
a) Distribution of year-on-year price changes for sticky and flexible HICPX items

(annual percentage changes)



b) Contribution of sticky and flexible items to NEIG and services inflation

(annual percentage changes and percentage point contributions)



Sources: Eurostat, Gautier et al. (2024) and ECB staff calculations.

Notes: “Sticky core” includes 39 HICPX items at the COICOP four-digit level and “Flexible core” includes 33. The weighted mean of sticky and flexible core inflation is based on the respective HICP weights. The median is calculated on the basis of the year-on-year inflation rates of all items in each subcategory and shaded areas represent the range of year-on-year rates (25th to 75th percentile). The vertical grey line in panel a) indicates peak HICPX inflation (March 2023). The latest observations are for September 2025.

Flexible-price goods drove the surge in NEIG inflation, while the persistence of services inflation reflects contributions from both sticky and flexible-price items (Chart B, panel b). For NEIG, the sharp increase in prices during the period 2021-23 was largely driven by flexible-price items, followed by a rapid decline that brought NEIG inflation back to subdued levels. NEIG inflation is currently relatively stable and moderate and predominantly driven by flexible-price items such as “garments” and “footwear”, while

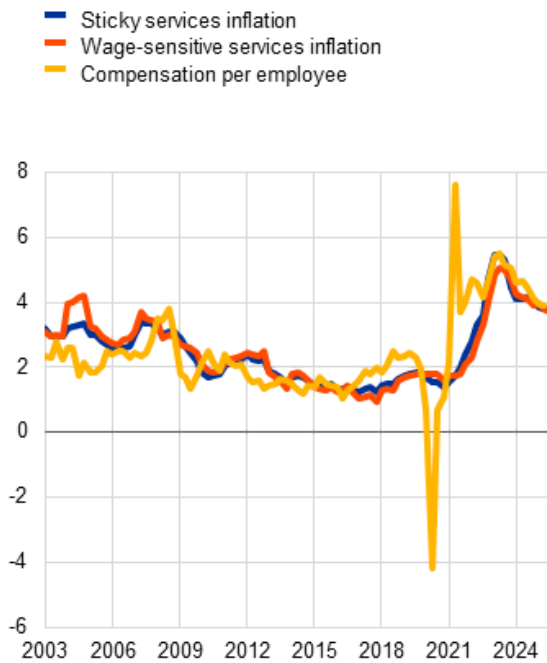
the contributions from sticky-price items are already back at their pre-pandemic average. By contrast, services inflation has remained elevated, reflecting the strong persistence of sticky-price items, whose contributions have declined only marginally since the beginning of this year (from 2.7 to 2.5 percentage points), while the contributions from flexible-price items have fallen more markedly (from 1.2 to 0.7 percentage points).^[4] The recent persistence of sticky-price inflation in services is consistent with Martínez Hernández et al. (2025), who found a more gradual pass-through of energy and supply-chain shocks to services than to NEIG inflation. Moreover, upward pressure from labour market shocks associated with strong wage growth has added to that persistence. Sticky services inflation thus co-moves strongly with wage dynamics and wage-sensitive services inflation (Chart C, panel a).

Chart C

Sticky services inflation, sticky and flexible core inflation and inflation expectations

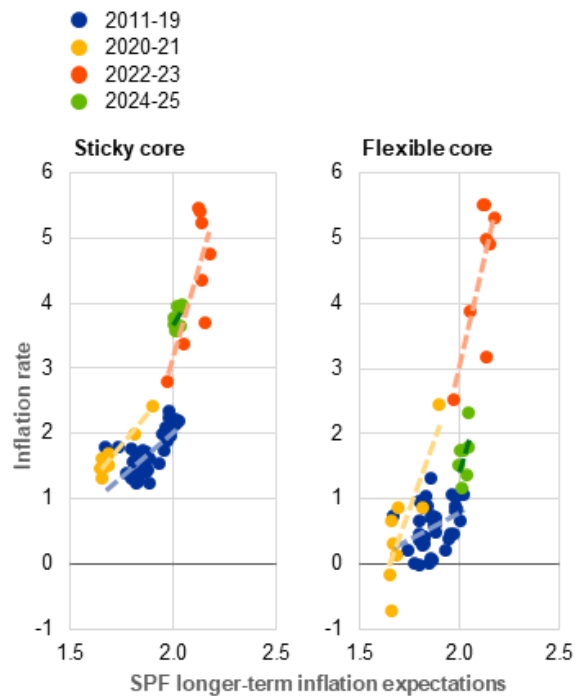
a) Sticky and wage-sensitive services inflation and compensation per employee

(annual percentage changes)



b) Sticky and flexible core inflation vs longer-term inflation expectations

(annual percentage changes)



Sources: Eurostat, Gautier et al. (2024), SPF and ECB staff calculations.

Notes: Panel a): chart based on quarterly data. Wage-sensitive services inflation is calculated as in Fagandini et al. (2024). Panel b): scatter plot based on quarterly data. Longer-term inflation expectations in the SPF may be as much as four or five calendar years in the future, depending on the timing of the survey. The latest observations in both panels are for the second quarter of 2025 for compensation per employee and the third quarter of 2025 for all other items.

Sticky core inflation in the euro area appears more closely linked to inflation expectations. A scatter plot of sticky and flexible core inflation against longer-term inflation expectations from the ECB's Survey of Professional Forecasters (SPF) shows that during the low-inflation period of 2011-19 sticky core inflation clustered tightly between 1% and 2.5%, close to SPF expectations (Chart C, panel b). By contrast, flexible core inflation was more dispersed, ranging from negative rates to well above 1%. This finding is reinforced

by estimates of quarterly New Keynesian Phillips curve specifications for each indicator, including its lagged value, the unemployment gap, inflation expectations (two years ahead and longer term) and import prices.^[5] Both medium and longer-term expectations significantly explain sticky core inflation, while they are insignificant for its flexible counterpart.

Taken together, the persistence of sticky core inflation underscores the roles of past cost shocks and elevated wage pressures. In view of both the moderation of the latter and anchored long-term inflation expectations, sticky core inflation is likely to disinflate further.

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1.

The Federal Reserve Bank of Atlanta regularly publishes a [sticky-price consumer price index \(CPI\)](#) for the United States. The Deutsche Bundesbank (2025) has also used a sticky-price indicator based on German CPI micro price data in the context of underlying inflation measures for Germany. Bodnár et al. (2025) show that items that are repriced only annually have remained an important factor behind the persistence of services inflation in the euro area.

2.

The data and replication files of Gautier et al. (2024) are available via [openICPSR](#). The authors compute frequency statistics at the COICOP five-digit level for the euro area using constant country weights (2017-

19 averages). On this basis, we derive euro area frequencies of price changes (including sales) by simple averaging at the COICOP four-digit level. In deriving the median frequency, we also apply the same authors' rule in considering only those items for which observations are available for at least three out of the four largest euro area countries (Germany, Spain, France and Italy). The remaining items are derived from the available countries.

3.

Out of the 39 sticky core items, eight are NEIG and 31 are services. Among the 33 flexible core items, 25 are NEIG and eight are services. Examples of sticky core items with relatively high weights are “restaurants, cafés and the like”, “maintenance and repair of personal transport equipment”, “social protection” and “other medical products, therapeutic appliances and equipment”. The largest flexible core items are “actual rentals for housing”, “motor cars”, “accommodation services”, “garments” and “furniture and furnishings”.

4.

A significant jump in flexible services inflation to 3.9% in April 2025 can be related to the exceptionally strong impact of this year's Easter period on prices of travel-related services (Eiglsperger et al., 2025).

5.

In particular, for each measure (sticky and flexible core) we estimate alternative specifications of a New Keynesian Phillips curve regression as follows: $\pi_t = \alpha + \beta\pi_{t-1} + \gamma Ugap_{t-1} + \rho IP_{t-2} + \delta\pi_t^e + \mu_t$, where π_t denotes the annualised quarterly rate of change in seasonally adjusted sticky or flexible core inflation, $Ugap_{t-1}$ is the unemployment gap lagged by one quarter, IP_{t-2} refers to the year-on-year growth rate of import prices and π_t^e represents inflation expectations.

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